

Postdoctoral Fellowships and Leadership

Fellowship vs. “Personal” Postdoc

What do you think are the differences?

Fellowship vs. “Personal” Postdoc

Fellowship:

- Research program builds on but substantially expands your thesis work in a new direction *not* driven by your thesis advisor.

Is this right for you?

- Have a plan to drive your own research program **without substantial oversight/guidance.**
- you already have an established set of collaborators.

Role of “Host Faculty” : They are a collaborator NOT the research lead or your boss. Mentorship, Admin PI, Ref. Letter.

- **Advantages:** Prestige, Salary, Research Budget, Visibility.
- **Down sides:** Expectations. It’s all on you – you are not part of a team.

Fellowship vs. “Personal” Postdoc

Personal Postdoc

- Directed research/work, designed by the PI. Usually with some % of independent research.

Is this right for you?

- Are you more interested in working in a collaboration/part of a team/mission?
- Can develop different technical skillsets – can lead to a industry or national lab position more readily (especially if you are plugging into a collaboration).
- You get more time to develop your own ideas with more guidance.

Role of the “Host Faculty” : They design the project and advise the research/project steps. They are your boss. Mentorship, Admin PI, Ref Letters

- **Advantages:** Pressure is less, you usually plug into a team, usually product/delivery oriented.
- **Down sides:** (can be) Less prestige and less time to do your own research.

What does it take to get a fellowship?

Bok Fellowship Selection Criteria: An example

- ~110 applicants
- After Round 1 ~ 20 applicants
- Round 1 – Letters were not used.
- Round 2 – Letters used to inform all of the criteria
- *caveat → Bok tends to go to someone already a postdoc and ready to be faculty soon.

What does it take to get a fellowship?

Bok Fellowship Selection Criteria: An example

- Categories: See [spread sheet](#).

Quality & Impact of Proposed Research

Quality & Impact of Past Research

Initiative

Fit with Institution

Components of the Rubric (Round 2)

Quality & Impact of Proposed Research

- **Importance of Proposal to the SubField:** Proposed Research Products are clearly articulated and strongly justified by candidate as being transformative for the Subfield (research should lead to significant results and/or novel tools critical to advancing the field)

Components of the Rubric (Round 2)

Quality & Impact of Proposed Research

- **Proposal as a Substantive Extension of Past Work:** Proposed Research is an Ambitious and Innovative Extension of Past Work (new directions that build on expertise, or next leap for an already successful program, Candidate is driven and absolutely ready to be an independent leader)

Components of the Rubric (Round 2)

Quality & Impact of Past Research

- **Creativity and Impact of Past Research** : Past Research is highly creative and has very strongly impacted the subfield (very high citation count evidenced/or expected relative to the standards of the subfield/nature of work, research statement conveys strong impact of candidate's work on advancing new research directions for the subfield - including leadership roles within e.g. collaborations/teams if applicable)

Components of the Rubric (Round 2)

Quality & Impact of Past Research

- **Productivity:** Past Research demonstrates very high productivity (papers, software development, grunt work and observing, instrumentation development/testing, code/pipeline development, Collaboration Leadership)

Components of the Rubric (Round 2)

Quality & Impact of Past Research

- **Leveraging of Available Resources:** Past Research demonstrates efficiently leveraging resources available to candidate (given institution resources)

Components of the Rubric (Round 2)

Initiative. [Leadership]

- **Scientific Initiative:** Candidate has provided evidence of very strong initiative in the field (Talks at conferences/Strong Communication Skills, working in teams, PI on proposals, Professional service - panels, referee on papers, Collaboration Service)

Components of the Rubric (Round 2)

Initiative [Leadership]

- **Mentorship Potential:** Candidate has strong record of mentoring graduate students or undergraduate students (e.g. publications led by students, advised multiple students, clear evidence of active mentorship of students)

Components of the Rubric (Round 2)

Initiative [Leadership]

- **Independence:** Past and Proposed research clearly illustrates the candidate's independent ideas

REFERENCE LETTERS MATTER HERE

Components of the Rubric (Round 2)

Fit With Institution

- **Interaction/Visibility at Steward:** Certain to be an active member of Steward Research Community and actively engage with graduate students, rather than isolated lone researcher (evidence of reaching out to form new collaborations, involvement in departmental activities/groups within and beyond subfield)

Components of the Rubric (Round 2)

Fit With Institution

- **Steward Resources (Facilities &/or Personnel):** Proposed Research Leverages Efficiently and Logically the resources available to the candidate, including strong discussion of logical connections with resources (faculty/telescopes/computing/data) at Steward.

Components of a Postdoc Application

- CV
- Cover Letter
- Research Statement: Past
- Research Statement: Proposed
- 3-4 Reference Letters

Components of a Postdoc Application

- CV → **Past Impact**, Productivity/Hustle, Mentorship, Service
- Cover Letter → FIT with Institution/Response to the Call.
- Research Statement: Past → **Demonstrated Impact**, Quality, Ambition, Creativity, Mentorship, Productivity
- Research Statement: Proposed → **Potential Impact**, Quality, Ambition, Creativity, Fit With Institution
- 3-4 Reference Letters → Leadership [Independence, Initiative], **Impact** → Should be Research Letters, ideally 1 not at your thesis institution]

Impact – Convince the committee that you have made impact and you will continue to have impact on the subfield.

- You can demonstrate this through:
 - Citations
 - Invited Talks & Accepted Conference talks.
 - Press Releases – Articles in the News
 - Proposals/Research/Missions/Other Papers informed by your work
 - Letter Writers
- The above doesn't just happen by posting a paper on the arxiv
 - GO to conferences – assess/enforce/build the impact of your work.
 - Build collaborations (obs and theorists - cross talk!)

Before we start looking at examples...

- <https://podcasts.apple.com/us/podcast/self-care-with-drs-sarah-jealousy/id1029803533?i=1000385390215>
- Self Care with Drs. Sarah Podcast. (Jealousy, Productivity)

- Sarah Rugheimer &
- Sarah Ballard

**Remember that
YOU ARE NOT HERE YET**



MAY 11, 2017 · 31 MIN LEFT

Self-care with Drs. Sarah: Jealousy
Self-care with Drs. Sarah

|| Pause

We're back for Season 2! Jealousy in the academic context is *rough.* The CV is a black hole from which nothing escapes, no matter how much gets accreted. Sarah B. should have a Gmail folder called "Friends Who Achieve Things That I'm Not Achieving", then maybe she'll actually read those emails. For more on this episode, please visit: <http://drssarahcare.tumblr.com/post/160568406290/episode-19-jealousy-were-back-the-sarabs>

CVs :

- **Junior:** Ekta Patel (UA → Miller), Kevin Wagner (UA → Sagan)
 - Press Releases
 - Conferences/Talks
 - Students Supervised
 - Proposals

CVs :

- **Junior:** Ekta Patel (UA → Miller), Kevin Wagner (UA → Sagan)
 - Press Releases
 - Conferences/Talks
 - Students Supervised
 - Proposals
- **Senior:**
 - Summary Section Leading
 - Proposals → More substantial
 - Student Supervised → More substantial
 - Papers and how to organize in terms of contributions

Example: Talks

When **Applied** as Grad Student

- G Besla: 22 total, 6 invited [Hubble]
- E Patel: 20 total, 6 invited [UA→Miller, NSF]
- N Garavito-Camargo: 15 total, 5 invited [UA →Flatiron Fellow]
- K Wagner: 13 total, 4 invited [UA → Sagan]
- S Scibelli: 22 total, 3 invited [UA → Jansky]
- D Setton: >11, 7 invited [UA ugrad → UPitt → Brinson Fellow]

Example: Talks

Applied as a **Postdoc**

- E Patel: 38 total, 23 invited [Hubble]
- N Garavito-Camargo: 47 total, 22 invited [Hubble]
- S Scibelli: 53 total, 18 invited [Sagan]
- D Setton: >29 total, 20 invited [Hubble]

Publication Records

- [Pepper et al. 2019](#)

We investigated the publication patterns of recipients of the NASA prize postdoctoral fellowships in the Hubble, Einstein, and Sagan programs from 2014 through 2017, using the NASA ADS reference system. We tabulated their publications at the point where fellowship applications were submitted, and we find that the 133 fellowship recipients in that time frame had a median of 6 +/- 2 first-author publications, and 14 +/- 6 co-authored publications. The full range of first author papers is **1 to 15, and for all papers ranges from 2 to 76**, indicating very diverse publication patterns.

Thus, while fellowship recipients generally have strong publication records, the distribution of both first-author and co-authored papers is quite broad; **there is no apparent threshold of publications necessary to obtain these fellowships.**

Example: Papers

Applied as Grad Student:

- G Besla : **3** first author from thesis (1 from ugrad). [Hubble]
- E Patel: **4** first author. [Miller, NSF]
- N Garavito-Camargo: **4** first author (1 from ugrad) [Flatiron Fellow]
- K Wagner: **7** first author from thesis (1 from ugrad) [Jansky]
- D Setton: **2** first author [Brinson Fellow]

Example: Papers – Co-authors

Applied as Grad Student:

- G Besla : **7** co-author. [Hubble]
- E Patel: **3** co-author. [Miller, NSF]
- N Garavito-Camargo: **6** co-author[Flatiron Fellow]
- K Wagner: **7** co-author [Jansky]
- D Setton: **5** co-author[Brinson Fellow]

Co-authoring papers is ALSO important → illustrates that you can work in and leverage collaborations, and/or can support students.